Introduction to Machine Learning for Economics
Final Assignment

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Part 1 – Supervised Learning

(i) Application and dataset

This exam project aims to predict the clinical label of autism spectrum disorder (Target variable: $y_i \in \{\text{ASD}, \text{Control}\}$). The data was collected on 33 sites in 28 different cities. For machine learning, this project uses a site-aware split so all subjects from a given site are kept together in either train or test. This prevents the model from learning site-specific artifacts. A stratified (by label) grouping keeps class balance similar across splits.

The data is from an international study by the Child Mind Institute as part of the International Neuroimaging Data-Sharing Initiative (INDI). The data is publicly available from the multi-site Autism Brain Imaging Data Exchange (ABIDE) and consists of 1,035 subjects. The number of subjects (871) shrink together with accuracy across all applied Machine Learning (ML) methods, when excluding images with lower image quality. Each subject has 200 brain regions of interest (ROIs) scanned at 176 time points. Producing 36,432,000.00 total observations. The dataset is almost perfectly balanced, as for every autistic person, the study found an age matched control, resulting in 51.2% control and 48.8% autistic subjects. In the next step, the correlations of the brain regions over time for each subject are calculated.

This functional connectivity between ROIs i and j is quantified by the Pearson correlation

$$r_{ij} = \frac{\sum_{t=1}^{T}(x_{it} - \bar{x}_i)(x_{jt} - \bar{x}_j)}{\sqrt{\sum_{t=1}^{T}(x_{it} - \bar{x}_i)^2}\,\sqrt{\sum_{t=1}^{T}(x_{jt} - \bar{x}_j)^2}}, \quad r_{ii} = 1. \tag{1}$$

Collecting all r_{ij} forms a symmetric correlation matrix $\mathbf{R} \in \mathbf{R}^{n\times n}$ with unit diagonal. Note that the shape of the matrix doesn't depend on time, as the correlations are summed over the time periods. Because $\mathbf{R}$ is symmetric and the diagonal is uninformative for between-ROI connectivity, only the strict upper triangle indices are retained, giving $p = \frac{n(n-1)}{2}$ unique connections (for $n = 200$, $p = 19{,}900$). Stacking these p values yields a subject-level feature vector $\mathbf{r} \in \mathbf{R}^p$.

Correlations are bounded ($r \in [-1, 1]$) and have skewed sampling distributions. Applying Fisher z-transformation, $z_{ij} = \frac{1}{2}\ln\left(\frac{1+r_{ij}}{1-r_{ij}}\right) = \operatorname{arctanh}(r_{ij})$, maps r to $\mathbb{R}$ and makes sampling distributions closer to normal with $\operatorname{Var}(z_{ij}) \approx \frac{1}{T_{\text{eff}}-3}$, where $T_{\text{eff}} \leq T$ accounts for temporal autocorrelation. This stabilizes variance and makes averaging across runs/subjects well-behaved; means are typically computed in z-space and optionally back-transformed via tanh. Because $\operatorname{arctanh}(\pm 1)$ diverges, r is limited to $[-1+\epsilon,\, 1-\epsilon]$ (e.g., $\epsilon = 10^{-6}$) before transforming.

With the ABIDE data set, the risk of overfitting is somewhat reduced, as the Pearson correlations between brain regions are complex as they retain a lot of information, hence the ML method doesn't necessarily find connections in the train data set that do not exist but may actually pick up on existing complex patterns.

(ii) Logistic Regression

Logistic regression models the positive–class probability via the logistic link

$$\Pr(y = 1 \mid \mathbf{x}) = \sigma(\eta) = \frac{1}{1 + \exp(-\eta)}, \qquad \eta = \beta_0 + \mathbf{x}^\top \boldsymbol{\beta},$$

higher values of η indicates stronger evidence in favor of $y = 1$ (ASD). The logistic function $\sigma(\eta)$ maps η scores to probabilities between zero and one. Taking the log-odds reveals why logistic regression is considered a linear classifier:

$$\log \frac{\Pr(y = 1 \mid \boldsymbol{x})}{\Pr(y = 0 \mid \boldsymbol{x})} = \beta_0 + \boldsymbol{\beta}^\top \boldsymbol{x}.$$

Thus, each coefficient β_j has a direct interpretation as the change in log-odds associated with a one-unit increase in feature x_j, holding all other features fixed. See coefficients on Pearson correlations in Appendix Table 2. Parameters are obtained by maximum likelihood, i.e., by minimising the negative log-likelihood

$$\mathcal{L}(\beta_0, \boldsymbol{\beta}) = -\sum_{i=1}^{n} \left[y_i \log \hat{p}_i + (1 - y_i) \log(1 - \hat{p}_i) \right], \quad \hat{p}_i = \sigma(\beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta})$$

[Hastie et al., 2009] [p. 119-121].
In the ABIDE application, the model learns η, which ranks subjects by ASD likelihood, and a threshold yields the ASD/Control decision. The standard logistic regression achieves essentially the same accuracy as the regularised ridge (L2) version ($\approx$ **67.7%** vs. **68.5%**). For the former case, find classification table 1, classification report 1, ROC curve 2, and Precision-Recall Curve 3, in the Appendix.

(iii) Support Vector Machines

As Support Vector Machines (SVM) reappear in question (viii) and (ix), it will be explained in more detail in the following. An SVM splits a space by a *hyperplane* such that the resulting two spaces separate observations into two categories: minus one and one (y_i). To split the observations in two, the hyperplane must find itself between minus one and one, hence it must equal zero: $y_i(w^T x_i + b) = 0$. Correct classification for all objects suggest: $y_i(w^T x_i + b) > 0$ as when transposed weights found by SVM times the explanatory variables of observation i plus the intercept found by SVM $(w^T x_i + b)$ exceeds zero, $\widehat{y_i}$ is classified as category one. If the observed y_i is category one as well then the SVM prediction was correct and consequently $y_i(w^T x_i + b) > 0$ holds. The reversed logic applies to the case of $(w^T x_i + b) < 0$, $\widehat{y_i} = -1$ and $y_i = -1$ then $y_i(w^T x_i + b) > 0$ is again satisfied and categorizes accurately.

SVM positions the hyperplane such that the smallest perpendicular distance of any training point to the boundary (the *margin*) is as large as possible. To calculate the perpendicular

distance, $y_i(w^T x_i + b)$ is divided by the square root of the sum of squared weights (euclidean norm): $\frac{y_i(w^T x_i + b)}{\|w\|}$, as this term is scale invariant and can be scaled such that the closest points satisfy $y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1$, this is advantageous for computer optimization because the distance between the two categories follows to be $\frac{1}{\|w\|} - \frac{-1}{\|w\|} = \frac{2}{\|w\|}$. As $\|w\|$ is in the denominator, maximizing the margin is equivalent to minimizing $\frac{1}{2}\|\mathbf{w}\|^2$. When the data are not perfectly separable and to prevent overfitting the train data set, SVM is applied with a *soft margin* with slack variables $\xi_i \geq 0$:

$$\min_{\mathbf{w},b} \ \tfrac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{N} \xi_i \quad \text{s.t.} \quad \xi_i \geq 0, \quad y_i(\mathbf{x_i}^\top \mathbf{w} + b) \geq 1 - \xi_i$$

(Hastie et al. [2009], p.420, eq. 12.8, whereas the original equation can be found in Cortes and Vapnik [1995], p.281, eq. 24). C trades off a wider margin against classification violations. Only points on or inside the margin (the *support vectors*) determine the solution. In the unconstrained formulation ξ_i can be defined as the hinge loss: $max[0, 1 - y_i(x_i^\top w + b)]$.

A linear SVM kernel has the advantage of coefficient extraction on the explanatory variables. Table 4 shows the beginning and the end of the 19,900.00 coefficients in descending order. The highest coefficient is 0.031 on the fisher z-transformation of the correlation across time between brain region 11 and 51 (Pearson correlation). The corresponding accuracy, with C=10 obtained from cross validation, is **69.54%**.

Non-linear boundaries are obtained via the kernel trick, e.g. the Gaussian Radial Basis Function (RBF): $K(\mathbf{x}, \mathbf{x}') = \exp(-\gamma\|\mathbf{x} - \mathbf{x}'\|^2)$ (Hastie et al. [2009], p. 424, eq. 12.21), with C and γ (kernel width) tuned by cross-validation.

In the ABIDE application, after cross validation, an RBF-SVM achieved **69.81%** test accuracy. RBF-SVM builds and factorizes an $n \times n$ kernel matrix, so training scales roughly between $\mathcal{O}(n^2)$ and $\mathcal{O}(n^3)$; grid-search over (C, γ) multiplies this cost. Hence, SVM can be computationally and time expensive. By performing a principal component analysis (pca) first, the computation time becomes minimal as the hyperplane is optimized in less dimensions. Performing the pca before running the RBF-SVM model increases test accuracy to **71.69%**, see classification table 4, classification report 3, ROC curve 5, and Precision-Recall Curve 6, in the Appendix.

(iv) Classification Trees

Classification and regression trees (CART) partition the feature space via axis-aligned splits, with hyperparameters such as tree depth and minimum leaf size serving as regularizers to control complexity. Pruning further reduces overfitting by removing branches that do not improve generalization on validation data. While CART offers high interpretability—allowing users to easily trace decision paths and understand feature importance—this comes at the cost of high variance, as small changes in the training data can lead to substantially different trees.

With cross validated parameters, the classification tree achieves **59.3%** accuracy in the ABIDE application. An image of the built tree can be seen in Figure 10. For further performance measures, see classification table 7, classification report 5, ROC curve 8, and Precision-Recall Curve 9, in the Appendix.

(v) Random Forests

Random Forests (RF) build an ensemble of decision trees (see the first produced tree in Appendix Figure 14) that are decorrelated through bootstrap sampling of the data and random selection of a subset of features at each split, thereby reducing the need for pruning individual trees (which are typically grown to full depth). In the ABIDE application, this decorrelation arises because each tree uses a random subset of features—the Pearson correlations between brain regions—at each branch node, resulting in trees that are less correlated with one another. Variable importance is estimated via permutation importance: the average decrease in Out-of-Bag (OOB) accuracy when a feature's values are randomly permuted across all trees (computationally heavy). Positive values indicate features that contribute useful information, while near-zero or negative values suggest limited or slightly detrimental impact in this fitted model (see Table 7 in the Appendix); however, these estimates should be interpreted with care, especially when features are highly correlated [Breiman, 2001].

In the ABIDE application, RF achieved an accuracy of **70.08%**. In the Appendix one can find the corresponding classification table 11, classification report 6, ROC curve 12, and Precision-Recall Curve 13. The Appendix also shows a discussion part arguing that RF's decorrelation attribute is particulary insightful in brain research, when respecting Panel structure (also related to subsection viii).

(vi) Gradient Boosting

Gradient Boosting builds a strong model by adding many small trees one after another: each new tree focuses on what the current model still gets wrong, i.e., it fits the remaining errors (more precisely, it takes a step in the direction of the negative gradient of the loss, so it is "functional gradient descent") [Friedman, 2001]. The learning rate controls how big each correction step is (small steps need more trees but often generalize better), while tree depth controls how complex each correction can be; together with the number of trees, they determine the bias–variance–runtime trade-off [Friedman, 2001, 2002]. As small steps, increased number in trees and deep trees can result in extensive run time. Gradient Boosting achieved **70.35%** accuracy. For further performance measures, see classification table 15, classification report 8, ROC curve 16, and Precision-Recall Curve 17, in the Appendix.

(vii) k-Nearest Neighbors (KNN)

KNN is a *memory-based* classifier: it does not fit a parametric model, but instead classifies a new subject by looking for the most similar subjects in the training set. Concretely, for a query point x_0 the computed distance is

$$d(i) = \|x_i - x_0\|,$$

pick the k closest training points $N_k(x_0)$, and then predict by a *majority vote* among these neighbors. If one codes the class as $y_i \in \{0, 1\}$, the local class probability is estimated by the neighborhood mean

$$\hat{p}(x_0) = \frac{1}{k} \sum_{x_i \in N_k(x_0)} y_i,$$

so predicting class 1 when $\hat{p}(x_0) > 0.5$ is exactly "majority vote" [Hastie et al., 2009, p. 15–16, Eq. (2.8)]. If one codes $y_i \in \{-1, +1\}$, the same rule can be written as $\hat{y}(x_0) = \text{sign}\Big(\sum_{x_i \in N_k(x_0)} y_i\Big)$. Since KNN relies on distances, it is *typically* recommended to standardize features when they are measured in different units [Hastie et al., 2009, p. 465]. In the ABIDE application, the predictors are Pearson correlations (hence already bounded in $[-1, 1]$ and on a common scale), so omitting standardization is defensible.

KNN achieved **63.61%** accuracy. For further performance measures, see classification table 18, classification report 9, ROC curve 19, and Precision-Recall Curve 20, in the Appendix.

(viii) Modeling panel data

The ABIDE data are longitudinal in the sense that each subject $i = 1, \ldots, 1035$ has 200 ROI time series observed over $T = 176$ fMRI time points, so outcomes may reflect both subject-specific heterogeneity and temporal dynamics. In the standard ABIDE pipeline used here, the time dimension is collapsed into a *static* functional connectivity (FC) representation by computing Pearson correlations between ROI time courses, see equation (1), producing one cross-sectional feature vector per subject. This transformation makes the supervised methods in (ii)–(vii) directly applicable. A complementary approach is to keep time variation and construct *dynamic* functional connectivity (dFC) features, e.g. via sliding-window correlations; ElNakieb et al. [2023] report that dFC improves accuracy relative to static FC (roughly +5 percentage points on average) and can reach state-of-the-art accuracy of **98.8%**, suggesting that temporal information can be highly informative for ASD prediction.

If the original panel structure is retained, model evaluation must remain *group-aware* (subject- and site-aware splits) to avoid leakage from repeated measurements of the same individual or scanner site. For logistic regression, longitudinal dependence can be handled by adding subject-specific fixed effects (intercepts) or random effects in a logistic mixed model, and inference should be clustered at the subject level to account for within-subject correlation [Wooldridge, 2010, Sec. 20.3.2]. For SVMs, one practical option is to fit a separate hyperplane for each time point t and then aggregate the resulting subject-level scores, e.g.

classify ASD if $\frac{1}{T}\sum_{t=1}^{T}\big(x_{i,t}^{\top}w_t + b_t\big) > 0$. This process resembles bag of time-instances: aggregating instance scores to a bag label is classic multiple-instance learning (MIL) as in Dietterich et al. [1997] or Maron and Lozano-Pérez [1998].Kittler et al. [1998] demonstrated how sum/mean rules are among the most effective and robust decision-combination schemes, an arithmetic expression closer to the expression above can be found in Alexandre et al. [2001], p. 1285, eq. 3. Imposing perfect classification in the ABIDE application suggests: $y_i(\frac{1}{176}\sum_{t=0}^{t=175} x_{i,t}^{\top}w_t + b_t) > 0$. Instead of averaging $x_{i,t}^{\top}w_t + b_t$, Platt [1998] scaling can provide probabilities, changing the decision threshold to 0.5 as $\frac{1}{1+exp(x_{i,t}^{\top}w_t+b_t)}$ maps from margin scores to probabilities. An alternative to the described late fusion process (van Hout et al. [2014], pages 2 and 3) is a single margin that learns in the augmented space $(\mathbf{x}, t)$. With a product kernel $K((\mathbf{x}, t), (\mathbf{x}', t')) = K_{\text{feat}}(\mathbf{x}, \mathbf{x}')\, K_{\text{time}}(t, t')$, the classifier is encouraged to be smooth over time while preserving feature similarities [Schölkopf and Smola, 2002, p.408],[Cristianini and Shawe-Taylor, 2000, p.45].

When coming to tree and boosting methods, an alternative to treating a subject as one static vector, is to summarize each subject's trajectory by many *time-interval features*. Concretely, for subject i and an interval $[a, b]$, compute a summary $z_{i,[a,b]}$ (e.g. mean, slope, variance, or a connectivity summary). A classification tree can then split a node by a simple threshold test $z_{i,[a,b]} \leq c$, where c is chosen to maximize impurity reduction as in CART Breiman et al. [1984]. If one additionally restrict candidate intervals to respect chronology (children intervals must lie within, or start after, the parent interval), then each root-to-leaf path becomes an explicit sequence of informative time segments. This is closely related to interval-based time-series trees such as the *Time Series Forest*, which builds trees from randomly selected intervals and chooses splits by standard criteria Deng et al. [2013]. Random Forests keep the same idea but average many decorrelated trees (bootstrap samples + random feature subsets), so one can use the same forest machinery while simply redefining "features" as the interval summaries $z_{i,[a,b]}$ Breiman [2001]. Gradient Boosting again uses trees as weak learners, but fits them sequentially to reduce a loss via the negative gradient; here, each weak learner is allowed to split on interval summaries instead of static covariates Friedman [2001, 2002]. For k-nearest neighbors one can apply the same representation trick (run kNN on the interval-feature vectors) or use a time-series distance such as Dynamic Time Warping before voting Berndt and Clifford [1994].

(ix) Semi-Supervised SVMs (S^3VMs)

Medical labels are costly, many unlabeled scans exist. In the ABIDE application for example, brain scans are very costly, many samples are not taken in the context of classifying autism, hence they are unlabeled, but they can still help in improving generalizability for existing models. The key intuition behind S^3VM is geometric: unlabeled points reveal where the data are dense in feature space, so a classifier should place its decision boundary in *sparse* regions—like putting a fence through empty space rather than cutting through a crowd. Transductive (or semi-supervised) SVMs implement exactly this idea by extending the usual maximum-margin

SVM: besides choosing (w, b) to separate the *labeled* data with a large margin, the method also chooses (explicitly or implicitly) labels for the *unlabeled* points so that the resulting margin is large on both labeled and unlabeled observations [Zhu, 2005, p. 13]. Bennett and Demiriz [1998][p. 369] describe this as learning from a labeled *training set* plus an unlabeled *working set*: each unlabeled point is allowed to be either class, and the optimization "picks" the label that yields the smaller slack/hinge penalty, so the hyperplane is encouraged to pass between unlabeled clusters rather than through them. Formally, this can be written as an SVM objective for labeled points plus an additional nonconvex penalty on unlabeled margins; with the aforementioned kernel trick, in the Reproducing Kernel Hilbert Space (RKHS) notation, one common view is

$$\min_f \sum_{i=1}^{\ell} \big(1 - y_i f(x_i)\big)_+ \; + \; \lambda_1 \|h\|_{\mathcal{H}_K}^2 \; + \; \lambda_2 \sum_{i=\ell+1}^{n} \big(1 - |f(x_i)|\big)_+,$$

where $(1 - |f(x)|)_+$ is a "hat-shaped" loss that penalizes *small absolute margins* on unlabeled points, making the boundary avoid dense unlabeled regions [Zhu, 2005, p. 15, eq. (3)]. The difficulty is that optimizing over unknown labels (or the hat loss) is nonconvex and the exact S^3VM solution is Non-deterministic Polynomial-time hard (NP-hard) [Zhu, 2005, p. 14]. Practical algorithms therefore use approximations, for example, the Concave–Convex Procedure (CCCP), which proposes that hat loss equals the sum of a convex and a concave function [Zhu, 2005, p.15]. Bennett and Demiriz [1998] show that the problem can be solved with mixed-integer programming when incorporating misclassification error constraints on both classes (-1 and 1) and adding a binary decision variable [Bennett and Demiriz, 1998, p.371, eq. 7]. However, when their algorithm was tested, it turned out to have difficulties handling a higher number of unlabeled data.

(x) SHAP: local/global explanations

SHAP explains a model prediction by treating the prediction as a "payout" that must be fairly split among the input features, using Shapley values from cooperative game theory [Shapley, 1953, p. 307]. For an instance x, start from a natural baseline $\phi_0 = \mathbb{E}[f(X)]$, i.e., what the fitted model predicts on average before any information about this specific x is revealed. Then, for any subset (coalition) S of features, define the coalition value as the model's expected output when only the coordinates in S are fixed to their values in x,

$$v(S) = \mathbb{E}[f(X) \mid X_S = x_S]\,.$$

Feature j receives credit for how much it increases the coalition value when it is "added" to a coalition, averaged over all possible coalitions (all orders of revealing features),

$$\phi_j = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!\,(|F| - |S| - 1)!}{|F|!} \Big(v(S \cup \{j\}) - v(S) \Big),$$

so the local explanation becomes the additive decomposition $f(x) \approx \phi_0 + \sum_{j=1}^{p} \phi_j$ [Lundberg and Lee, 2017, pp. 4768–4777]. This construction is intuitive: if revealing feature j typically moves the model output a lot (relative to the baseline), then ϕ_j is large; if revealing j rarely changes the output, then ϕ_j is near zero. "Global" importance is obtained by aggregating local attributions across observations, e.g., by reporting $\frac{1}{n}\sum_{i=1}^{n} |\phi_{ij}|$ as a dataset-level importance score. In practice, exact Shapley averaging is combinatorial, so SHAP relies on efficient estimators (e.g., KernelSHAP) and, for tree ensembles, fast exact algorithms such as TreeSHAP [Lundberg et al., 2020, p. 56]. Key caveats follow directly from the definition of $v(S)$: if features are strongly correlated, different choices of conditioning (conditional vs. marginal "interventional" expectations) can split credit in unintuitive ways [Aas et al., 2021]; if the background distribution used to approximate $\mathbb{E}[\cdot]$ changes (e.g., different reference samples), then ϕ_0 and all ϕ_j can shift; and SHAP explains the fitted function $f(\cdot)$, not causal effects, so it should be read as "what the model used," not "what truly causes the outcome." Sensible baselines, correlation-aware variants, and stability checks across folds help mitigate these issues.

Part 2 – Unsupervised Learning

(i) Clustering with constraints

The core idea of clustering is to group observations so that points within the same cluster are similar, while points in different clusters are dissimilar. This can be done either by *optimising* an explicit objective (e.g. k-means), or by *greedy merging/splitting* without a global objective, as in hierarchical clustering (agglomerative clustering repeatedly merges the closest clusters under a linkage criterion) [Hastie et al., 2009, Ch. 14]. In k-means, "similar" is measured by squared Euclidean distance to a cluster center, so the algorithm chooses assignments and centers to minimize within-cluster dispersion:

$$\min_{\{\mu_k\},\{z_{ik}\}} \sum_{i=1}^{n} \sum_{k=1}^{K} z_{ik} \, \|x_i - \mu_k\|^2, \quad z_{ik} \in \{0,1\}, \quad \sum_{k=1}^{K} z_{ik} = 1$$

[Hastie et al., 2009, Ch. 14]. In practice, geometry alone can produce implausible groupings, so constraints are added that represent knowledge beyond distances. Must-link and cannot-link constraints encode pairwise relations: must-link says two observations must be in the same

cluster, cannot-link says they must not. With assignment variables, this can be written as

$$(i,j) \in M \Rightarrow z_{ik} = z_{jk}\ \forall k, \qquad (i,j) \in C \Rightarrow \sum_{k=1}^{K} z_{ik} z_{jk} = 0.$$

Intuitively, these constraints restrict the assignment step: even if a point is close to a center, assignment might be forbidden if it contradicts prior knowledge. COP-k-means implements this by accepting a reassignment only if all constraints remain feasible [Wagstaff et al., 2001]. More generally, constraints can be imposed *hard* (feasibility) or *soft* (penalties), so the method trades off compactness against violations [Basu et al., 2004]. Balance constraints add a different type of information: they do not say who should go together, but rather how many observations each cluster may contain, e.g. via

$$L_k \leq \sum_{i=1}^{n} z_{ik} \leq U_k, \quad k = 1, \ldots, K$$

so each cluster must contain at least L_k and at most U_k points. Conceptually this turns clustering (including constrained variants of k-means or hierarchical merging) into a constrained allocation problem; algorithmically it is often handled by casting the assignment step as a linear/integer optimisation or min-cost flow subproblem [Bradley et al., 2000, de Maeyer et al., 2023].

(ii) Three evaluation metrics

If ground-truth labels are not available, one can evaluate whether clusters are tight and well separated (internal validation). The silhouette coefficient does this pointwise. Let $a(i)$ be the average distance from point i to its own cluster (cohesion), and $b(i)$ the smallest average distance from i to any other cluster (separation). Then

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \in [-1, 1].$$

This definition is intuitive: if $a(i) \ll b(i)$, then i is much closer to its own cluster than to the next-best cluster, so $s(i)$ is near 1; if $a(i) \approx b(i)$, then i sits on the boundary and $s(i) \approx 0$; if $a(i) > b(i)$, then it is arguably "misclustered" and $s(i) < 0$ [Rousseeuw, 1987]

If ground-truth labels exist, an external metric is preferable. The Adjusted Rand Index (ARI) compares two partitions by counting pairwise agreements (together vs. apart) and then adjusting for the agreement expected by chance; this makes ARI harder to inflate by trivial clusterings [Hubert and Arabie, 1985]. A third common internal metric is the Calinski–Harabasz index, which formalizes "between-cluster separation relative to within-cluster spread":

$$\mathrm{CH}(K) = \frac{B_K/(K-1)}{W_K/(n-K)},$$

where B_K is between-cluster dispersion and W_K within-cluster dispersion; larger values indicate more separated and tighter clusters [Caliński and Harabasz, 1974].

Part 3 – Dimensionality Reduction

(i) ICA vs. PCA and MDS (pros/cons)

Dimensionality reduction replaces high-dimensional descriptions by a smaller number of informative coordinates. PCA does this by keeping directions of maximal variance: the first component solves $\max_{\|v\|=1} \mathrm{Var}(Xv)$ (and later components add orthogonality), so PCA is excellent for linear compression and noise reduction [Hastie et al., 2009, ch. 14]. Its limitation is that it only targets second-moment structure: it removes linear correlation but does not explicitly separate latent sources.

Independent Component Analysis (ICA) changes the goal from "uncorrelated" to "independent." In the linear mixing model $x = As$, ICA seeks an unmixing matrix W so that $\hat{s} = Wx$ has statistically independent components. The intuition is that independence is stronger than decorrelation, and ICA can exploit non-Gaussianity (higher-order information) to separate mixed signals [Hyvärinen and Oja, 2000]. The tradeoff is that ICA relies on stronger assumptions (approximate independence, non-Gaussian sources) and has interpretation ambiguities (ordering/sign/scale).

Multidimensional Scaling (MDS) starts from pairwise dissimilarities d_{ij} (not necessarily raw features) and finds low-dimensional points y_i whose distances match d_{ij} as well as possible, e.g. by minimizing a stress objective

$$\min_{\{y_i\}} \sum_{i<j} w_{ij} \big(\|y_i - y_j\| - d_{ij}\big)^2.$$

This is useful when "distance" is meaningful but coordinates are not (or are high-dimensional), but results depend strongly on the chosen dissimilarity and the scaling of the stress criterion [Borg and Groenen, 2005, Kruskal, 1964].

(ii) Visualizing dynamic data

For dynamic data, the goal is not only to map observations, but to show how they move over time. A simple way to use standard DR methodologies is to convert each time point (or time window) into a feature vector $z_{i,t}$ (e.g., window means, slopes, or windowed correlations), then apply PCA/ICA/MDS to the pooled set $\{z_{i,t}\}$ and plot the resulting low-dimensional coordinates as trajectories over t. The plot becomes immediately interpretable: similar dynamics trace similar paths, and regime shifts appear as sharp turns. In an MDS setup, one can define distances between time windows and embed (i, t) points so that temporal evolution is visible in the geometry; in practice, mild smoothness can also be imposed by discouraging large jumps between $y_{i,t}$ and $y_{i,t-1}$ to reflect continuity in time [Borg and Groenen, 2005].

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Appendix: Evaluation Metrics and Plots

In the following the content of the majority of tables and figures in the Appendix will be explained.

Classification table (confusion matrix)

For a binary task with labels $y \in \{-1, +1\}$ and predictions $\hat{y} \in \{-1, +1\}$, the *classification table* counts outcomes

	$\hat{y} = -1$	$\hat{y} = +1$
$y = -1$	TN	FP
$y = +1$	FN	TP

$$n = \mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}.$$

Overall accuracy is the share of correct decisions,

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{n}.$$

Sensitivity (true positive rate) and specificity (true negative rate) are

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \qquad \text{TNR} = \frac{\text{TN}}{\text{TN} + \text{FP}},$$

and the false positive rate is $\text{FPR} = 1 - \text{TNR} = \text{FP}/(\text{FP} + \text{TN})$.

Classification report

The report lists, for each class, *precision*, *recall*, and *F1* together with the *support* (class size). The definitions are

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \qquad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \qquad \text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

Two aggregates are commonly reported: the *macro average*, which is the unweighted mean of the per-class scores, and the *weighted average*, which weights each class by its support. Accuracy coincides with $\frac{\text{TP}+\text{TN}}{n}$ from the classification table.

ROC curve and AUC

Many classifiers output a continuous score $s(x)$. Thresholding s produces different confusion matrices and hence different (FPR, TPR) pairs. The *ROC curve* traces

$$\big(\ \text{FPR}(\tau),\ \text{TPR}(\tau)\ \big) \quad \text{as the threshold } \tau \text{ varies,}$$

with FPR on the horizontal axis and TPR on the vertical axis. The *area under the ROC curve* (AUC) summarizes ranking quality over all thresholds; $\text{AUC} = 0.5$ indicates random guessing, and larger values indicate better discrimination. Because it depends on ranks, ROC/AUC is relatively insensitive to class imbalance.

Precision–Recall curve

The *PR curve* plots precision against recall when the decision threshold varies,

$$\big(\ \text{Recall}(\tau),\ \text{Precision}(\tau)\ \big).$$

This view is particularly informative for imbalanced problems because it focuses on the quality of positive predictions (especially important in classifing autism because a FP can have drastic impact on an individual): as recall increases, precision typically decreases, and the curve

visualizes that trade-off. Looking at the mathematical definition, this tradeoff is characterized by FN decreasing while FP increases. So the model get's better in avoiding classifying FN but that comes with the backdrop of classifying more FP. A common single-number summary is either the maximum F1 achieved across thresholds or the area under the PR curve.

Reading the figures

The confusion matrix reports counts by actual class (rows) and predicted class (columns). The classification report converts these counts into precision, recall, and F1 per class, along with macro and support-weighted averages. The ROC curve and its AUC evaluate threshold-free ranking performance, while the PR curve emphasizes positive-class quality under class imbalance.

Logisic Regression

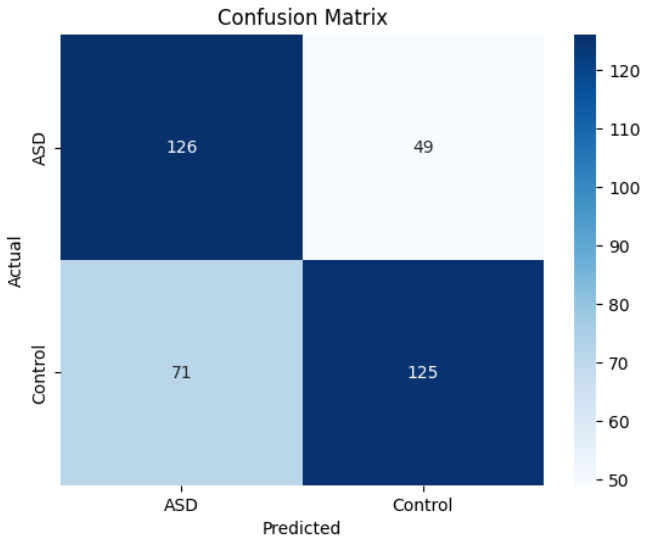

Figure 1: Classification Table

Class	Precision	Recall	F1	Support
ASD	0.640	0.720	0.677	175
Control	0.718	0.638	0.676	196
Accuracy			0.677	371
Macro avg	0.679	0.679	0.677	371
Weighted avg	0.681	0.677	0.676	371

Table 1: Classification report

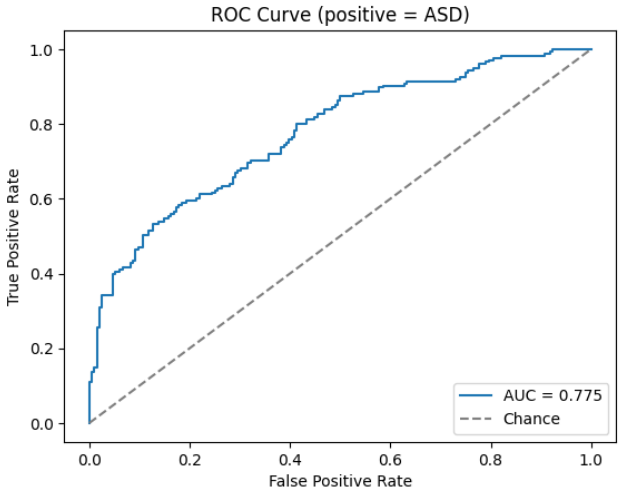

Figure 2: ROC Curve

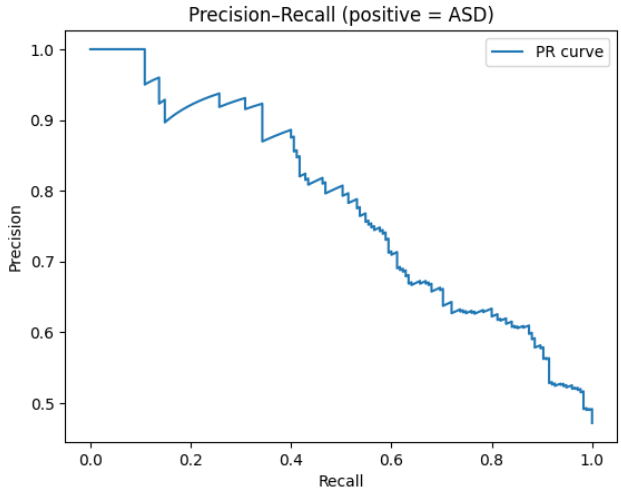

Figure 3: Precision-Recall Curve

Index	Feature	Coefficient
5444	fc_roi_030__roi_110	0.1343
1984	fc_roi_011__roi_051	0.1303
1965	fc_roi_011__roi_032	0.1278
15995	fc_roi_112__roi_124	0.1267
6567	fc_roi_037__roi_071	0.1262
2154	fc_roi_012__roi_033	0.1248
7314	fc_roi_041__roi_176	0.1244
18103	fc_roi_140__roi_174	0.1221
6955	fc_roi_039__roi_136	0.1220
3314	fc_roi_018__roi_086	0.1218
	...	
12767	fc_roi_081__roi_089	-0.1238
11570	fc_roi_071__roi_127	-0.1278
12055	fc_roi_075__roi_106	-0.1294
15655	fc_roi_108__roi_142	-0.1317
8481	fc_roi_049__roi_107	-0.1395
6881	fc_roi_039__roi_062	-0.1491

Table 2: Top and bottom logistic regression coefficients (sorted by magnitude)

Support Vector Machines

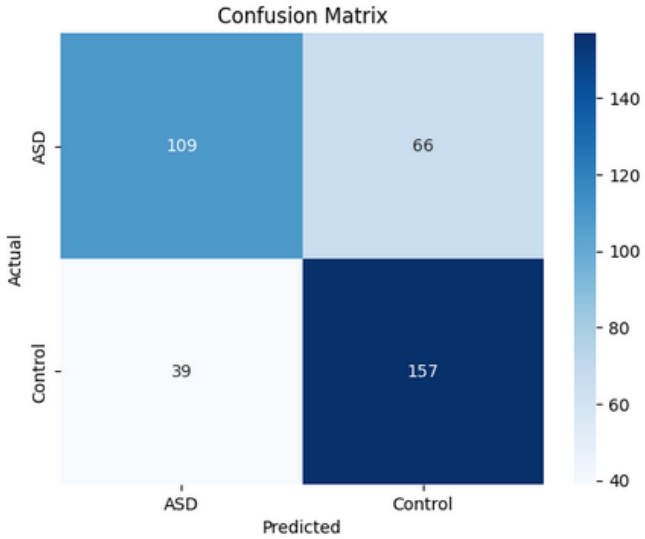

Figure 4: Classification Table

Class	Precision	Recall	F1	Support
ASD	0.736	0.623	0.675	175
Control	0.704	0.801	0.749	196
Accuracy			0.717	371
Macro avg	0.720	0.712	0.712	371
Weighted avg	0.719	0.717	0.714	371

Table 3: Classification Report

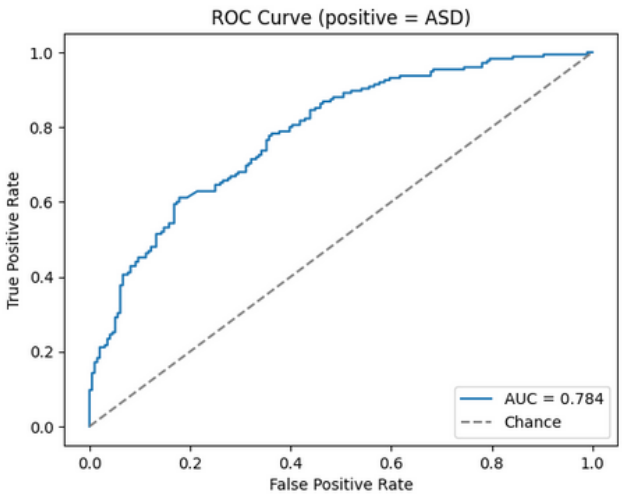

Figure 5: ROC Curve

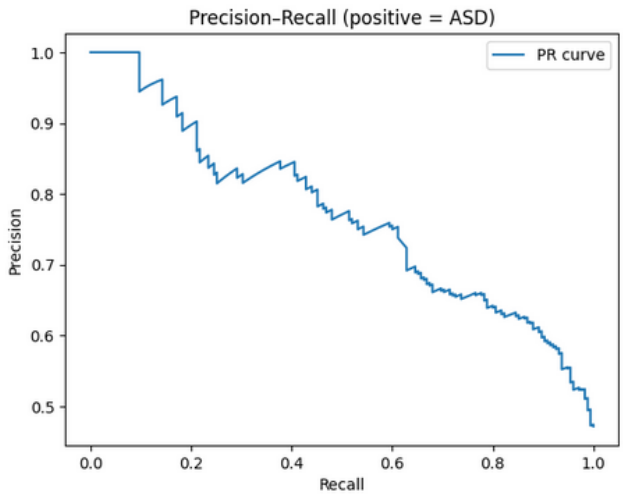

Figure 6: Precision-Recall Curve

Index	Feature	Weight
1984	fc_roi_011__roi_051	0.030925
5444	fc_roi_030__roi_110	0.028715
15995	fc_roi_112__roi_124	0.028223
6567	fc_roi_037__roi_071	0.028218
14800	fc_roi_099__roi_151	0.026790
	...	
16961	fc_roi_123__roi_188	-0.026266
1625	fc_roi_009__roi_071	-0.027616
11570	fc_roi_071__roi_127	-0.029110

Table 4: Top and bottom SVM (linear) feature weights

Classification Trees

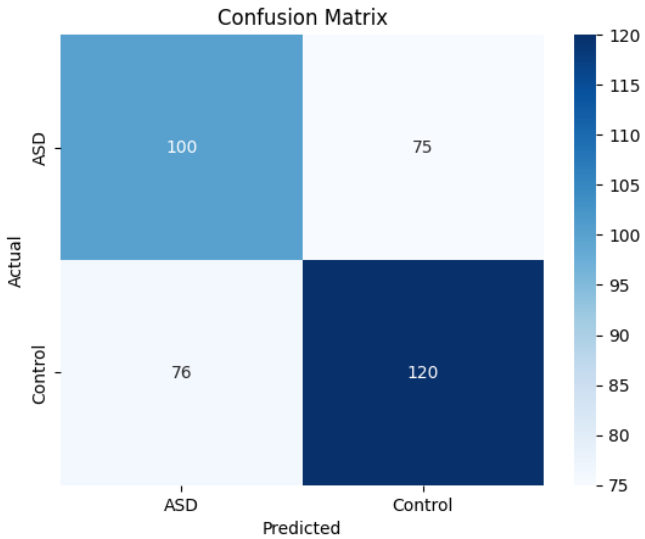

Figure 7: Classification Table

Class	Precision	Recall	F1	Support
ASD	0.568	0.571	0.570	175
Control	0.615	0.612	0.614	196
Accuracy			0.593	371
Macro avg	0.592	0.592	0.592	371
Weighted avg	0.593	0.593	0.593	371

Table 5: Classification Report

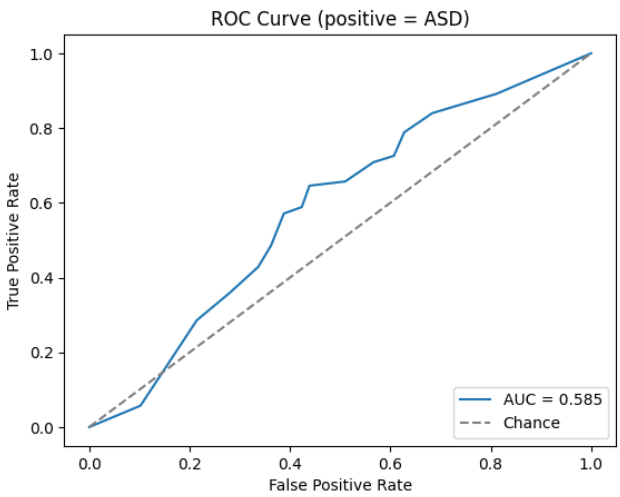

Figure 8: ROC Curve

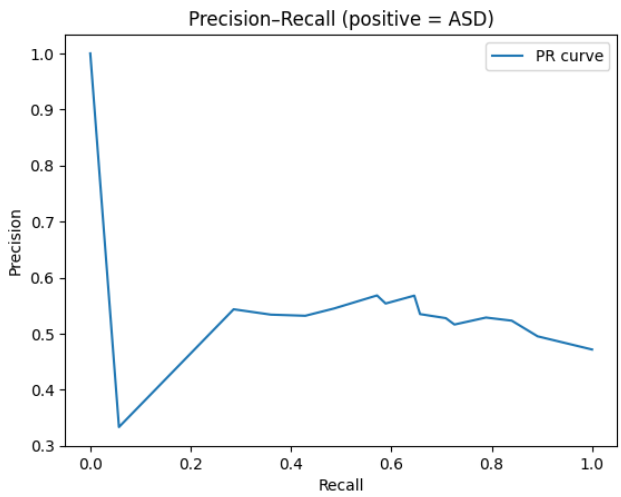

Figure 9: Precision-Recall Curve

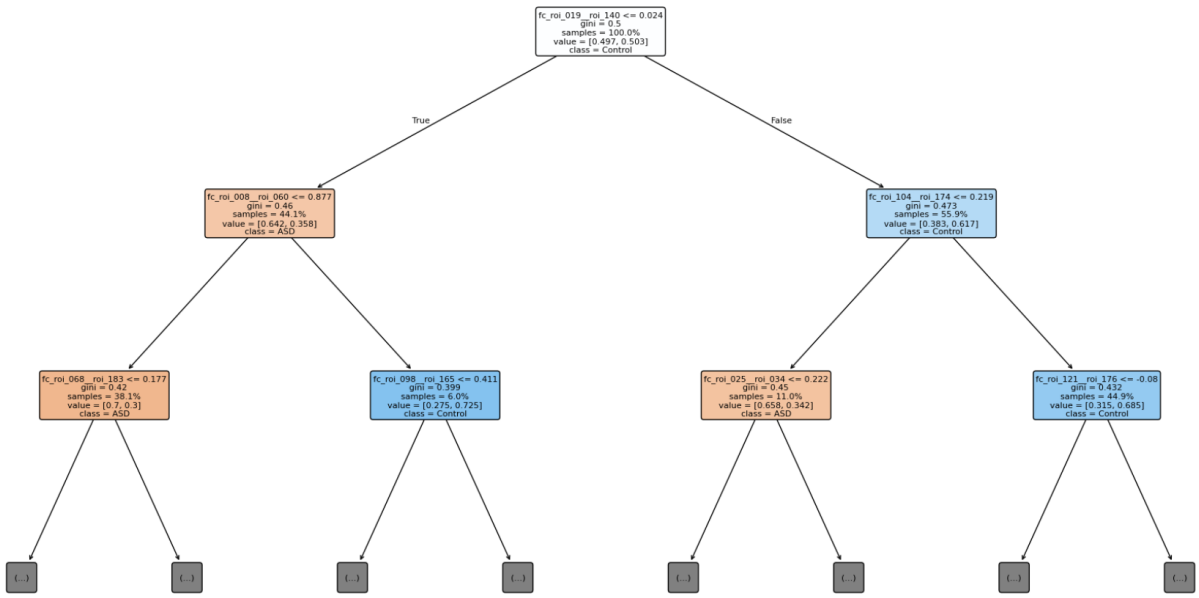

Figure 10: Classification Tree

Random Forests

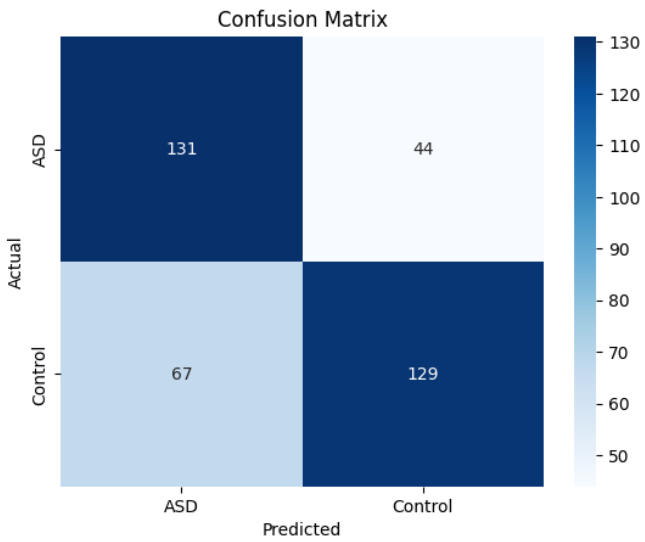

Figure 11: Classification Table

Class	Precision	Recall	F1	Support
ASD	0.662	0.749	0.702	175
Control	0.746	0.658	0.699	196
Accuracy			0.701	371
Macro avg	0.704	0.703	0.701	371
Weighted avg	0.706	0.701	0.701	371

Table 6: Classification Report

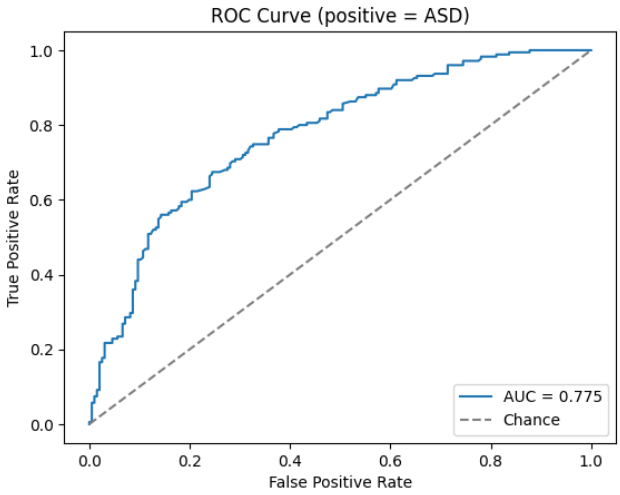

Figure 12: ROC Curve

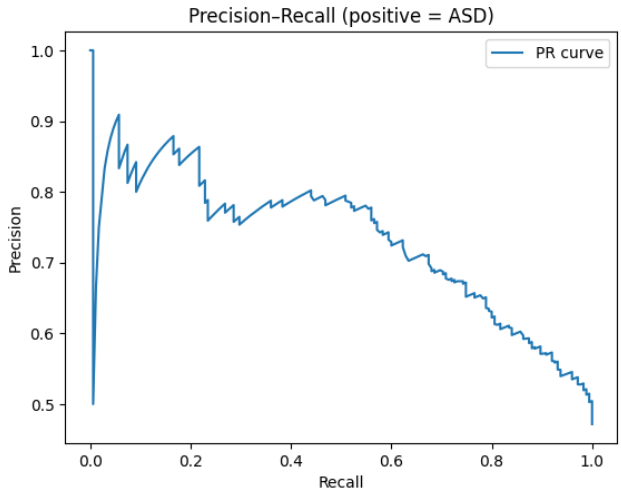

Figure 13: Precision-Recall Curve

In the context of brain research, RF in Panel data has a special advantageous feature, as the decorrelated trees could have the potential of finding the sequence of brain regions, repeated chains of synaptical connections pass through. In more detail, if collective stimuli exceed a certain threshold, a trigger results in a specific chain of synaptical connections. With many triggers, this chain may repeat. The repeated chain passes through specific brain regions, whereas those brain regions have different functionalities, hence, it heavily depends on the kind of stimuli, to determine which brain regions will be affected by the chain of synaptical connections. Liu et al. [2021] point out that in the future, task-based (different stimuli) fMRI have the potential to improve the diagnosis of ASD. Long story short, specific stimuli affect specific brain regions, and that is where decorrelation is from advantage, as RF could have the

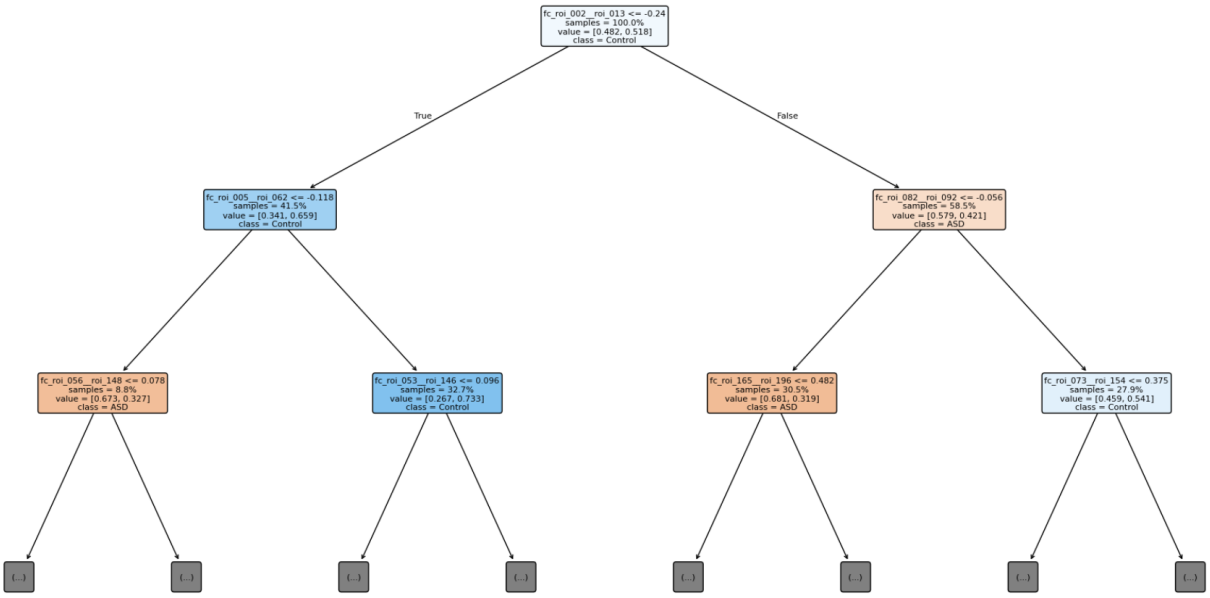

Figure 14: Classification Tree

Feature	**Importance (mean)**	**Importance (std)**
fc_roi_166__roi_191	0.005121	0.000809
fc_roi_028__roi_059	0.004852	0.001078
fc_roi_139__roi_140	0.004582	0.001235
fc_roi_064__roi_105	0.004582	0.001235
fc_roi_019__roi_140	0.004582	0.005121
...	...	...
fc_roi_022__roi_112	-0.003774	0.002156
fc_roi_046__roi_193	-0.004043	0.002173
fc_roi_005__roi_140	-0.004852	0.002017

Table 7: Permutation importance (top and bottom features)

potential to find the specific sequences of brain regions, isolated from other brain regions. This can give additional valuable inside into the origins of psychological disorders, such as autism in the ABIDE data set. This can give a deeper insight into disorders resulting from possible under or over repeated triggers, which could deliver some explanation to common findings in the ABIDE dataset that connect autism to reduced functional coherence of long-range intra-hemispheric cortico-cortical default mode circuitry and increase in local and short-range cortico-subcortical coherence.

Gradient Boosting

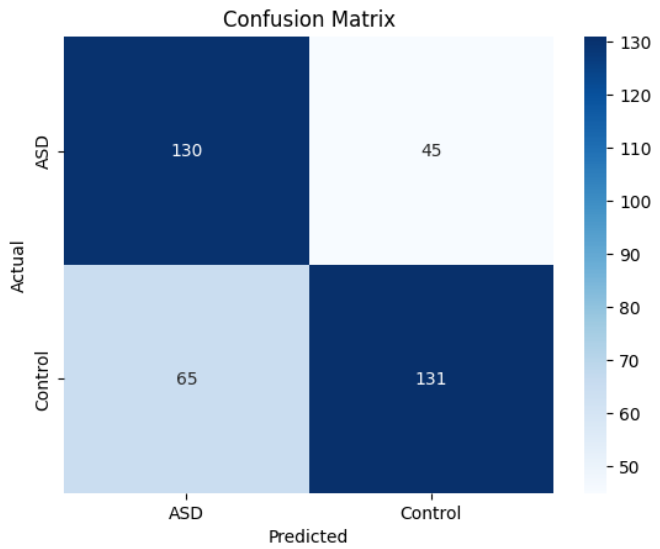

Figure 15: Classification Table

Class	Precision	Recall	F1	Support
ASD	0.667	0.743	0.703	175
Control	0.744	0.668	0.704	196
Accuracy			0.704	371
Macro avg	0.705	0.706	0.704	371
Weighted avg	0.708	0.704	0.704	371

Table 8: Classification Report

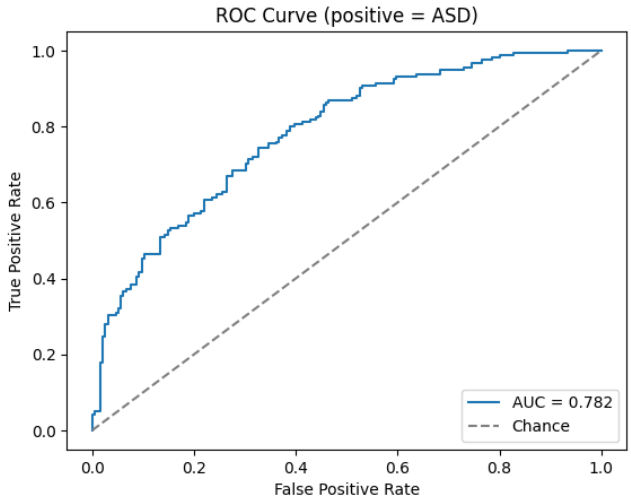

Figure 16: ROC Curve

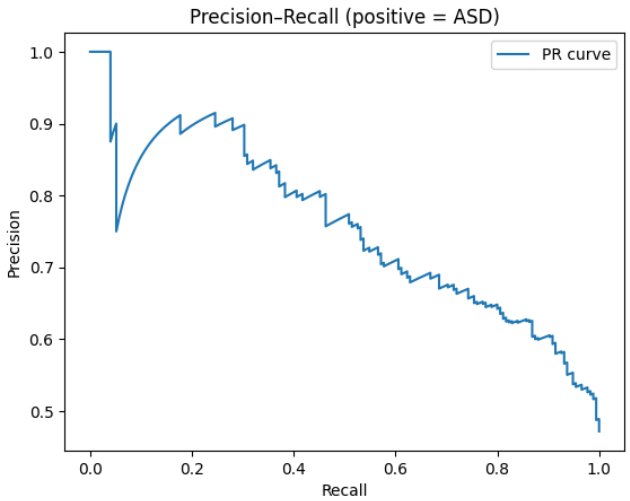

Figure 17: Precision-Recall Curve

k-Nearest Neighbors

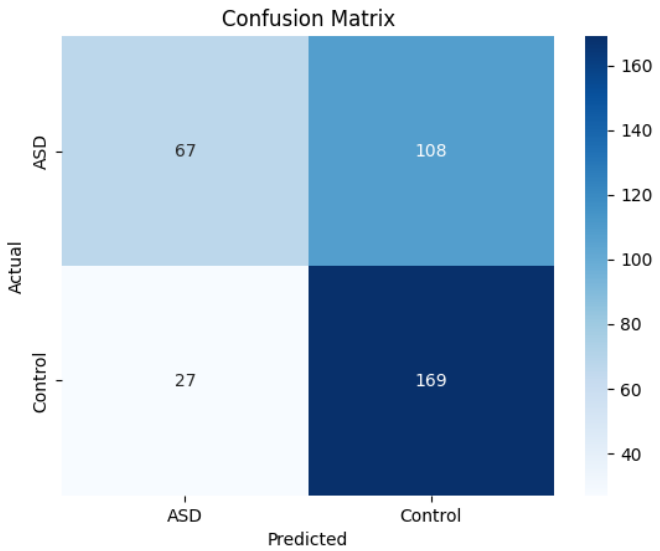

Figure 18: Classification Table

Class	Precision	Recall	F1	Support
ASD	0.713	0.383	0.498	175
Control	0.610	0.862	0.715	196
Accuracy			0.636	371
Macro avg	0.661	0.623	0.606	371
Weighted avg	0.659	0.636	0.612	371

Table 9: Classification Report

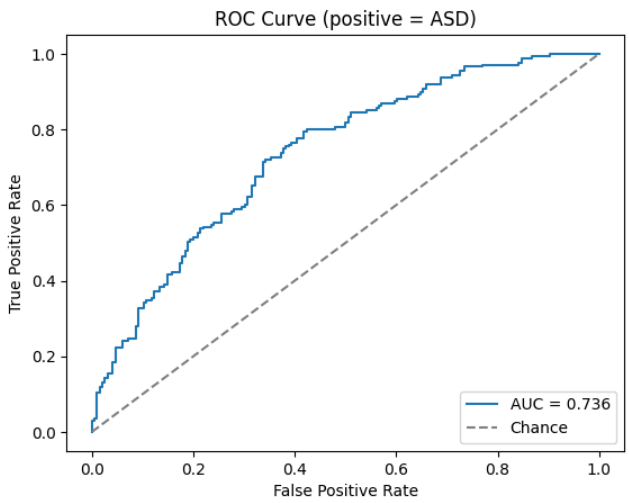

Figure 19: ROC Curve

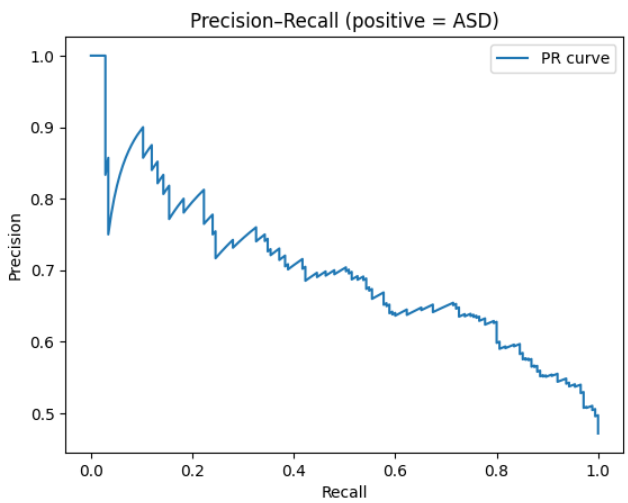

Figure 20: Precision-Recall Curve